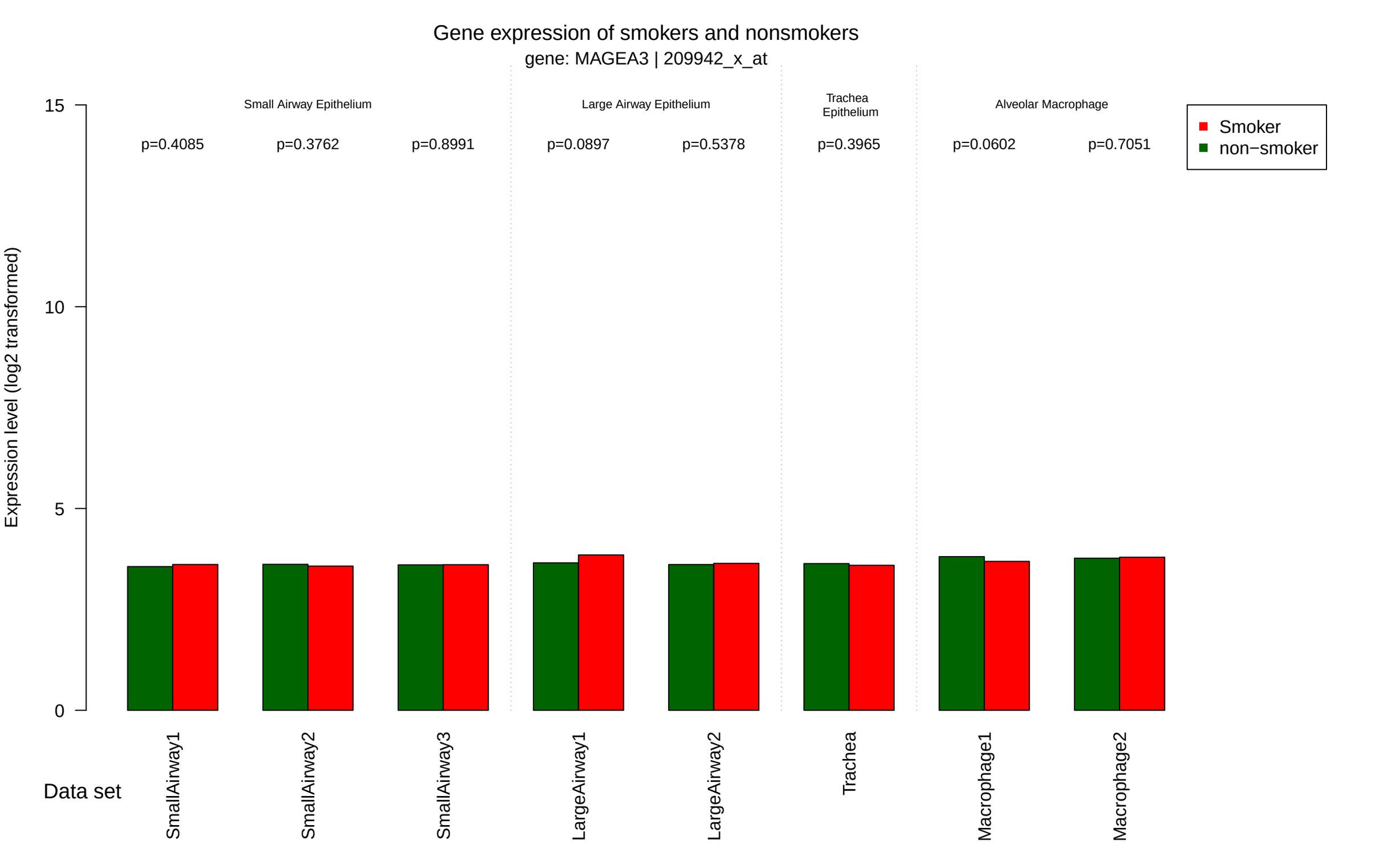
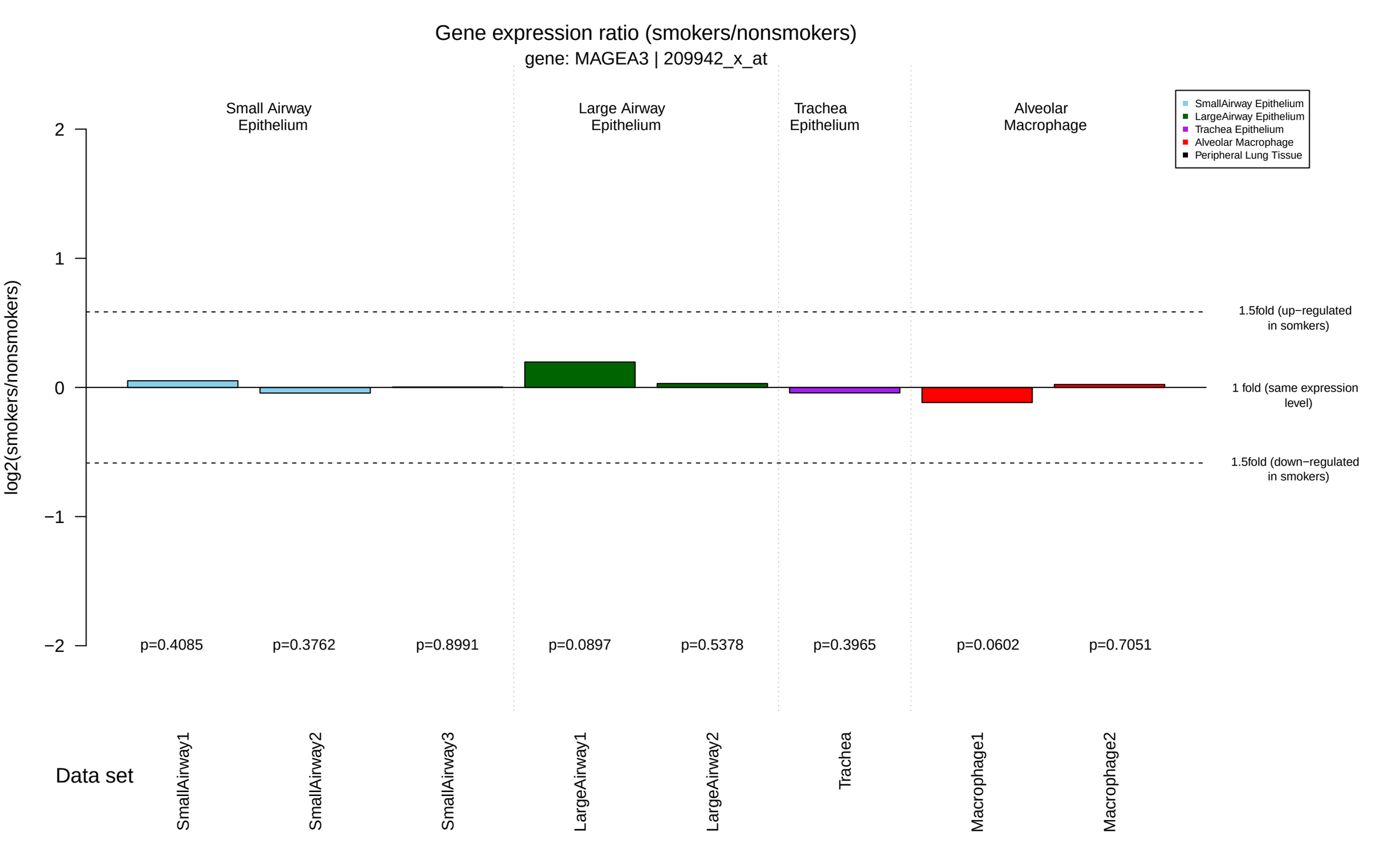
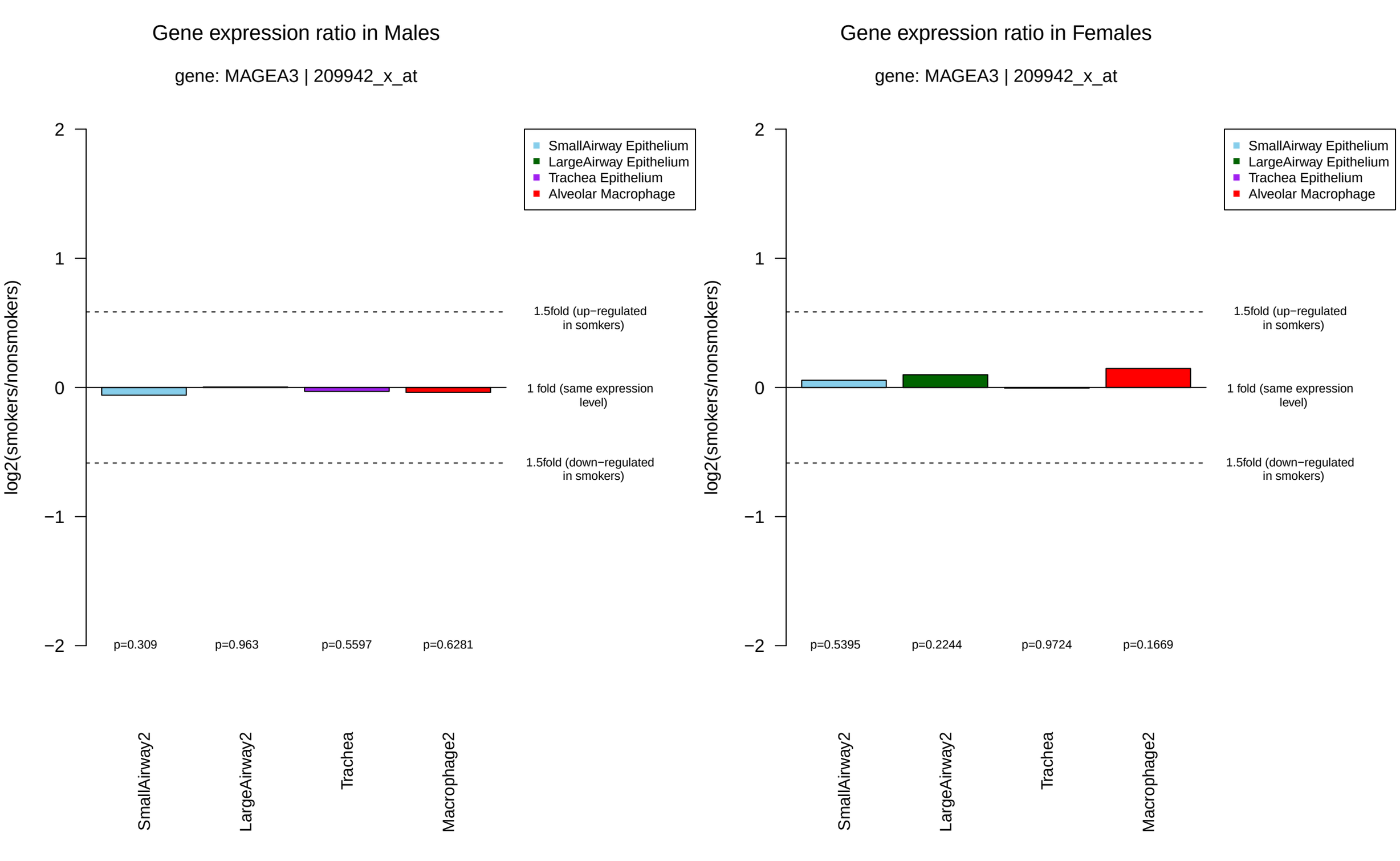
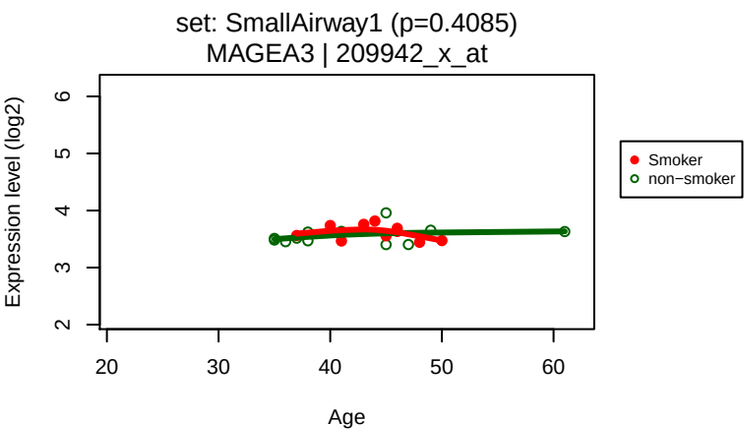
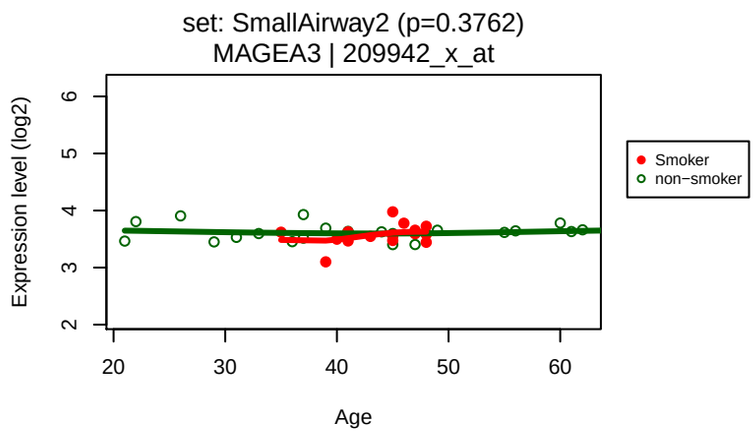
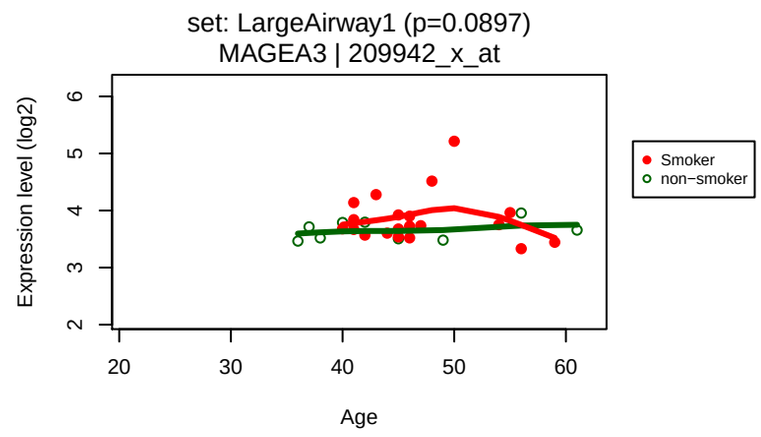

Gene expression of smokers and nonsmokers
gene: MAGEA3 | 209942_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4085
p=0.3762
p=0.8991
p=0.0897
p=0.5378
p=0.3965
p=0.0602
p=0.7051
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

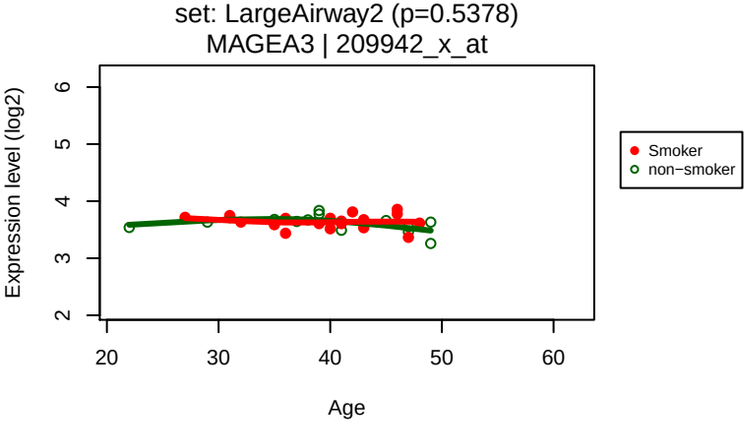

Gene expression ratio (smokers/nonsmokers)
gene: MAGEA3 | 209942_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4085
p=0.3762
p=0.8991
p=0.0897
p=0.5378
p=0.3965
p=0.0602
p=0.7051
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

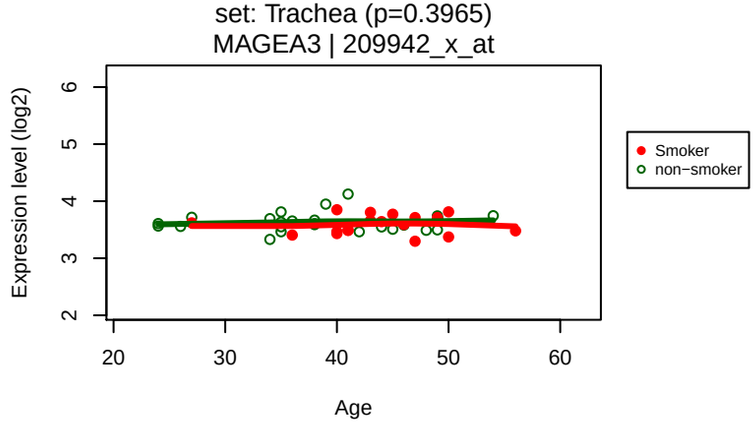

Gene expression ratio in Males
gene: MAGEA3 | 209942_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.309
p=0.963
p=0.5597
p=0.6281
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAGEA3 | 209942_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5395
p=0.2244
p=0.9724
p=0.1669
SmallAirway2
LargeAirway2
Trachea
Macrophage2

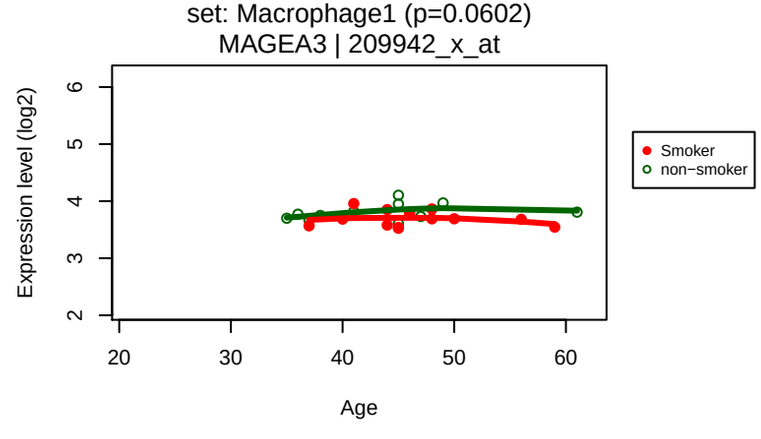

set: SmallAirway1 (p=0.4085)
MAGEA3 | 209942_x_at
Expression level (log2)
Age
Smoker
non−smoker

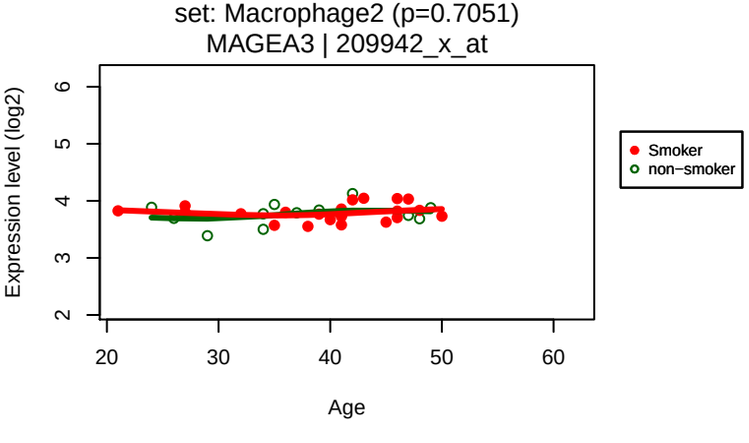

set: SmallAirway2 (p=0.3762)
MAGEA3 | 209942_x_at
Expression level (log2)
Age
Smoker
non−smoker

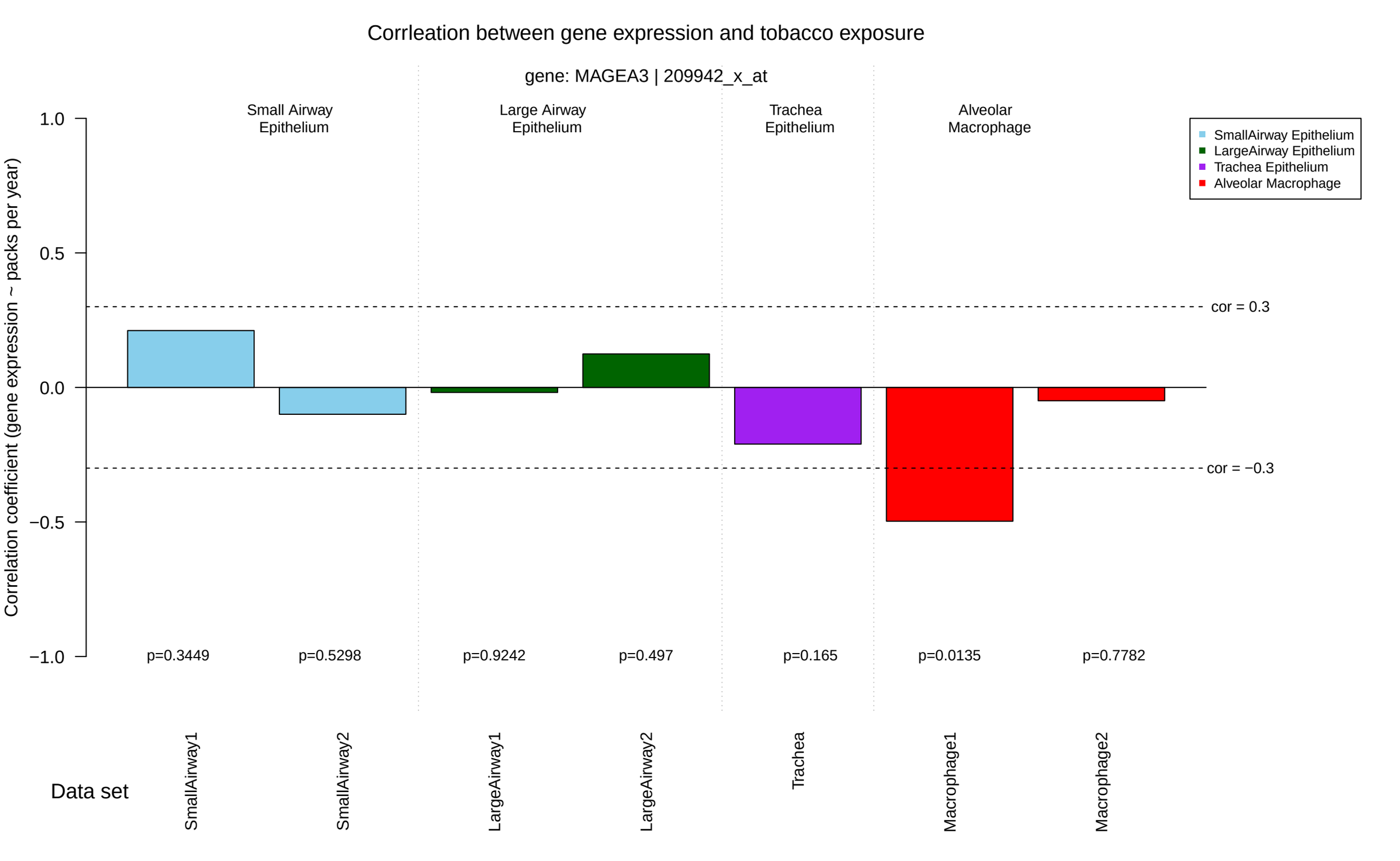

set: LargeAirway1 (p=0.0897)
MAGEA3 | 209942_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5378)
MAGEA3 | 209942_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3965)
MAGEA3 | 209942_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0602)
MAGEA3 | 209942_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7051)
MAGEA3 | 209942_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAGEA3 | 209942_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3449
p=0.5298
p=0.9242
p=0.497
p=0.165
p=0.0135
p=0.7782
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2